Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	P.Swetha nagasri
Student Name	Kallu shresta priya
Roll Number	2520030127
Branch	CSE
Course Outcome	CO-2

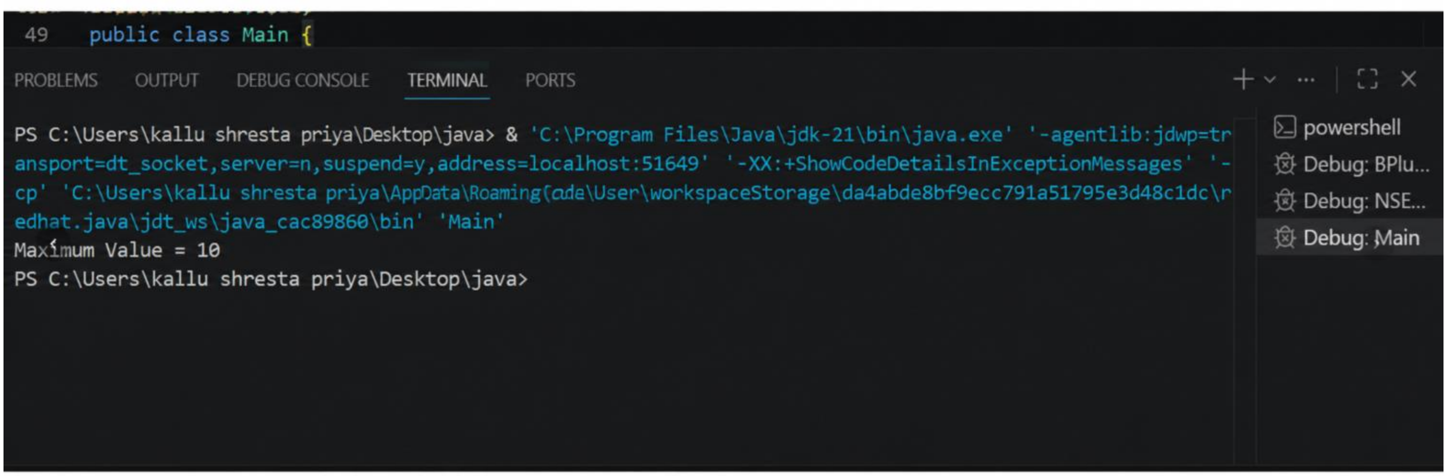
Case Study Topic: Range Update Queries using Segment Tree with Lazy Propagation

Work Done Summary:
In this case study, a Segment Tree with Lazy Propagation was implemented to efficiently perform range update operations on an array. Lazy Propagation postpones updates to child nodes until they are required, thereby reducing unnecessary computations. This approach allows range updates and queries to be executed in logarithmic time, making it suitable for large datasets and real-time applications.

Implementation Details:

- Created a Segment Tree to store information about array segments.
- Created a Lazy array to store pending updates.
- Implemented range update functionality.
- Checked for No Overlap, Partial Overlap, and Complete Overlap conditions.
- Applied updates directly for complete overlap.
- Stored pending updates in the Lazy array.
- Propagated pending updates to child nodes when required.
- Updated parent nodes after processing child nodes.
- Verified logarithmic complexity for update operations.

Result: Screenshots / Output



Time Complexity

- Segment Tree Construction: O(n)

- Range Update with Lazy Propagation: O(log n)
- Range Que

GitHub Link:

Student Signature	Faculty Signature